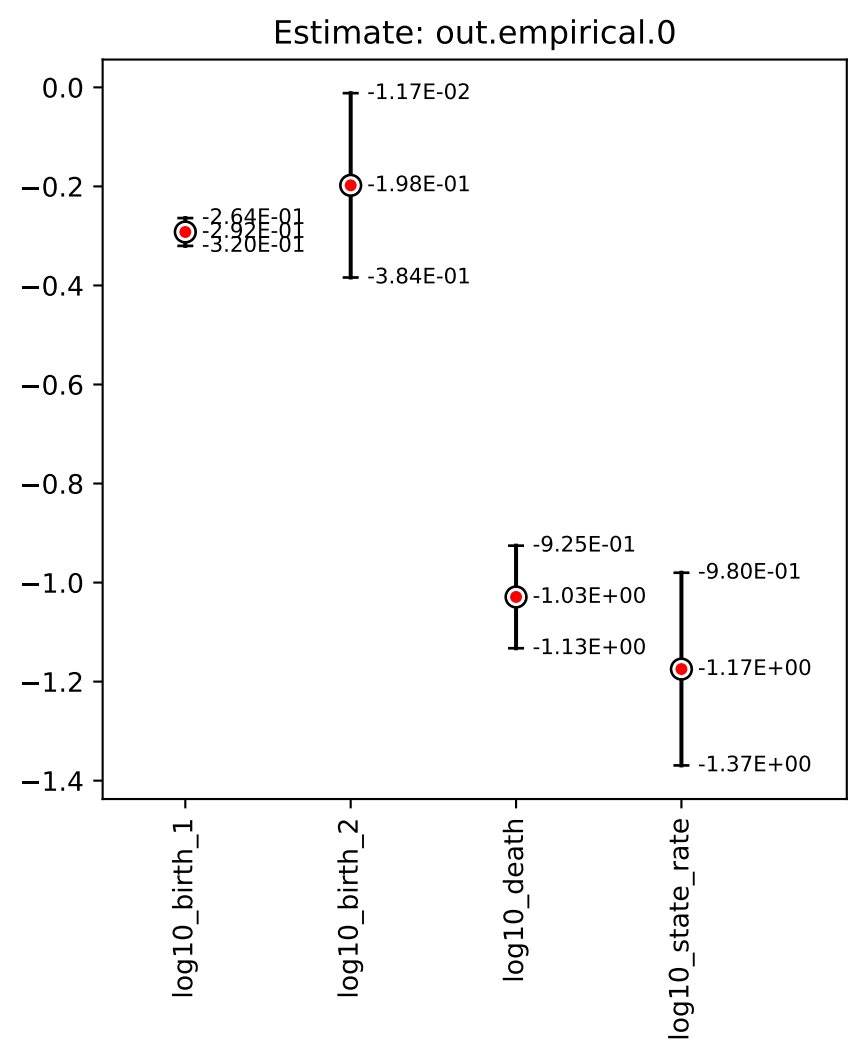
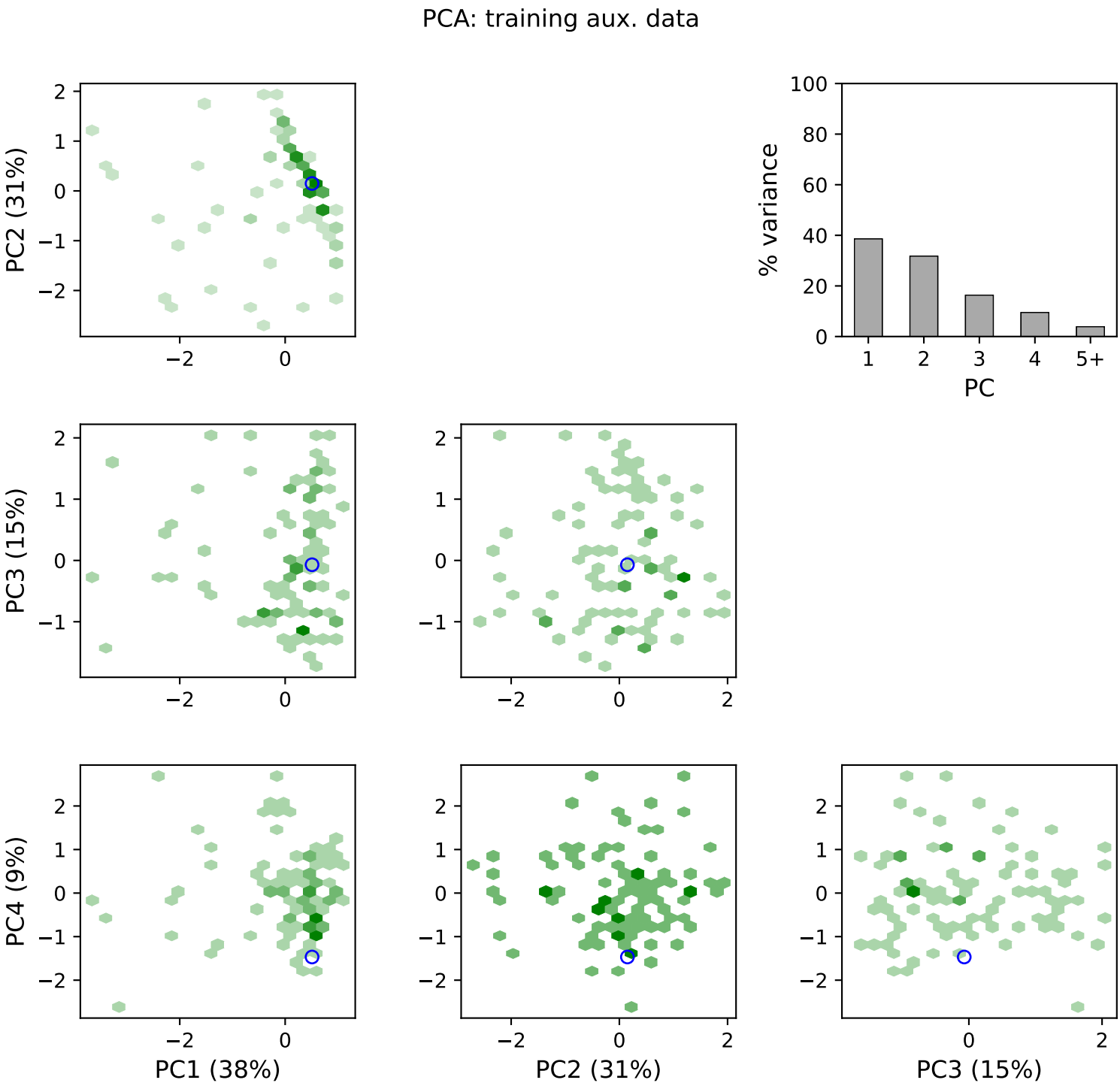
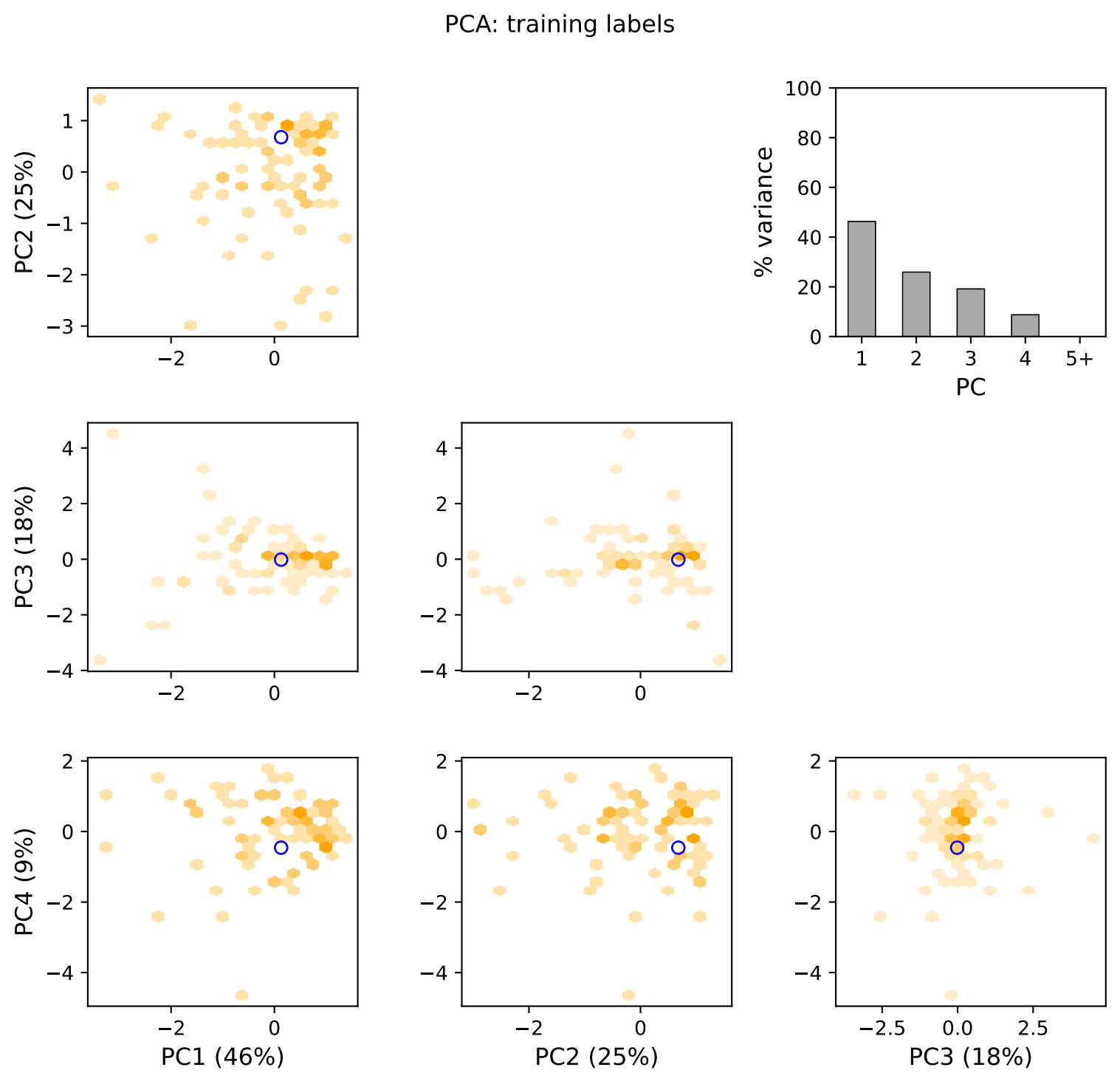
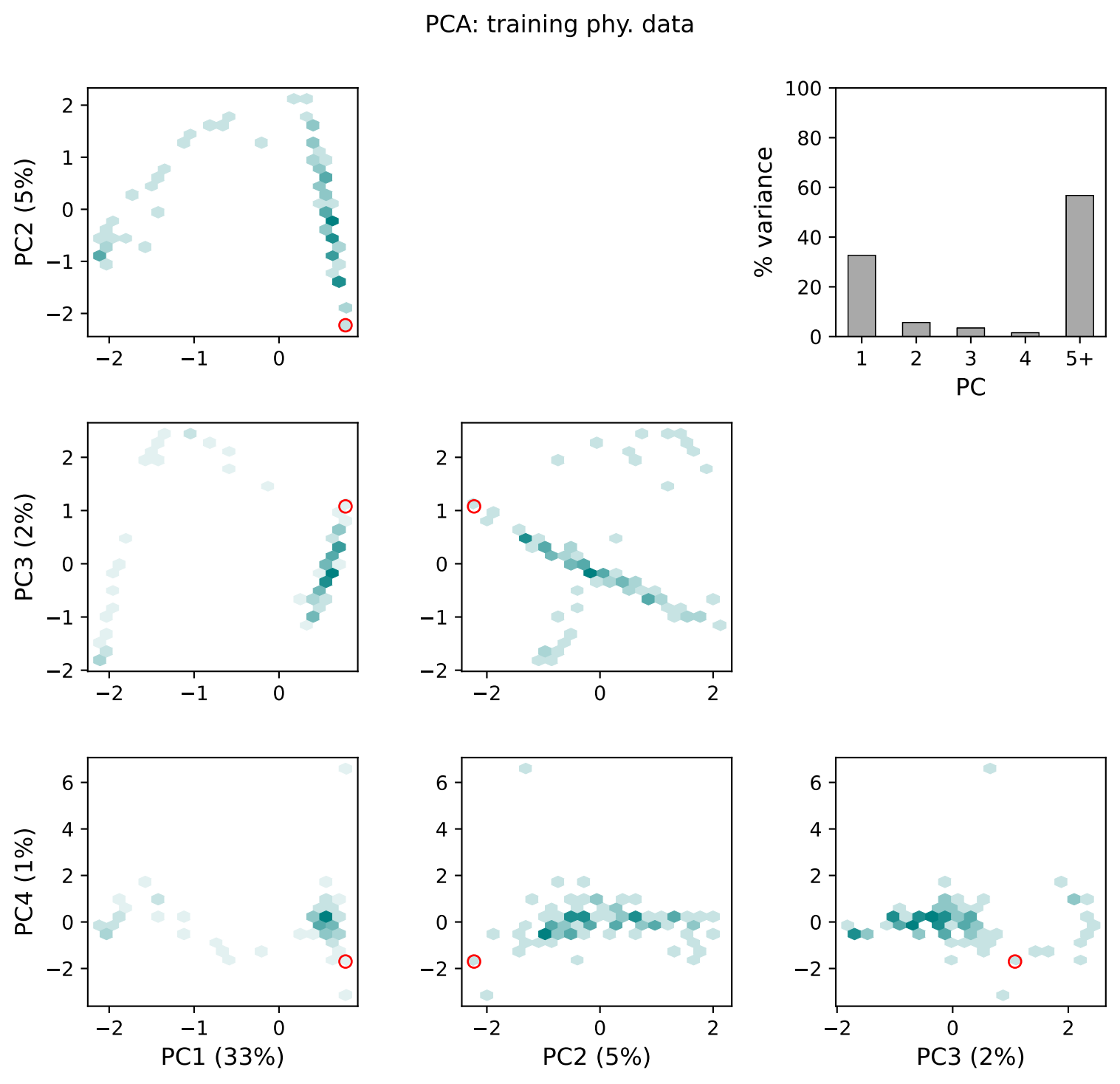
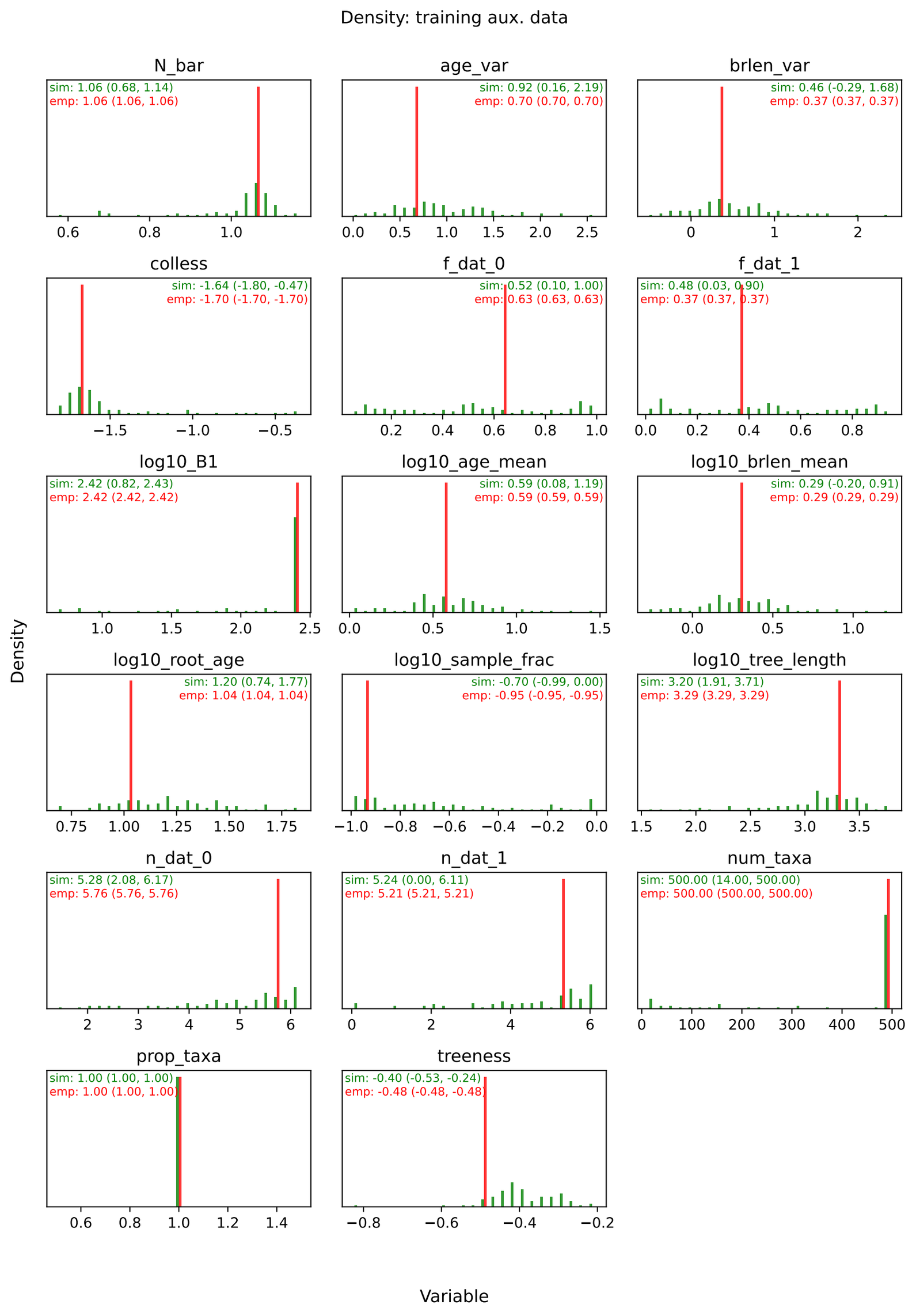
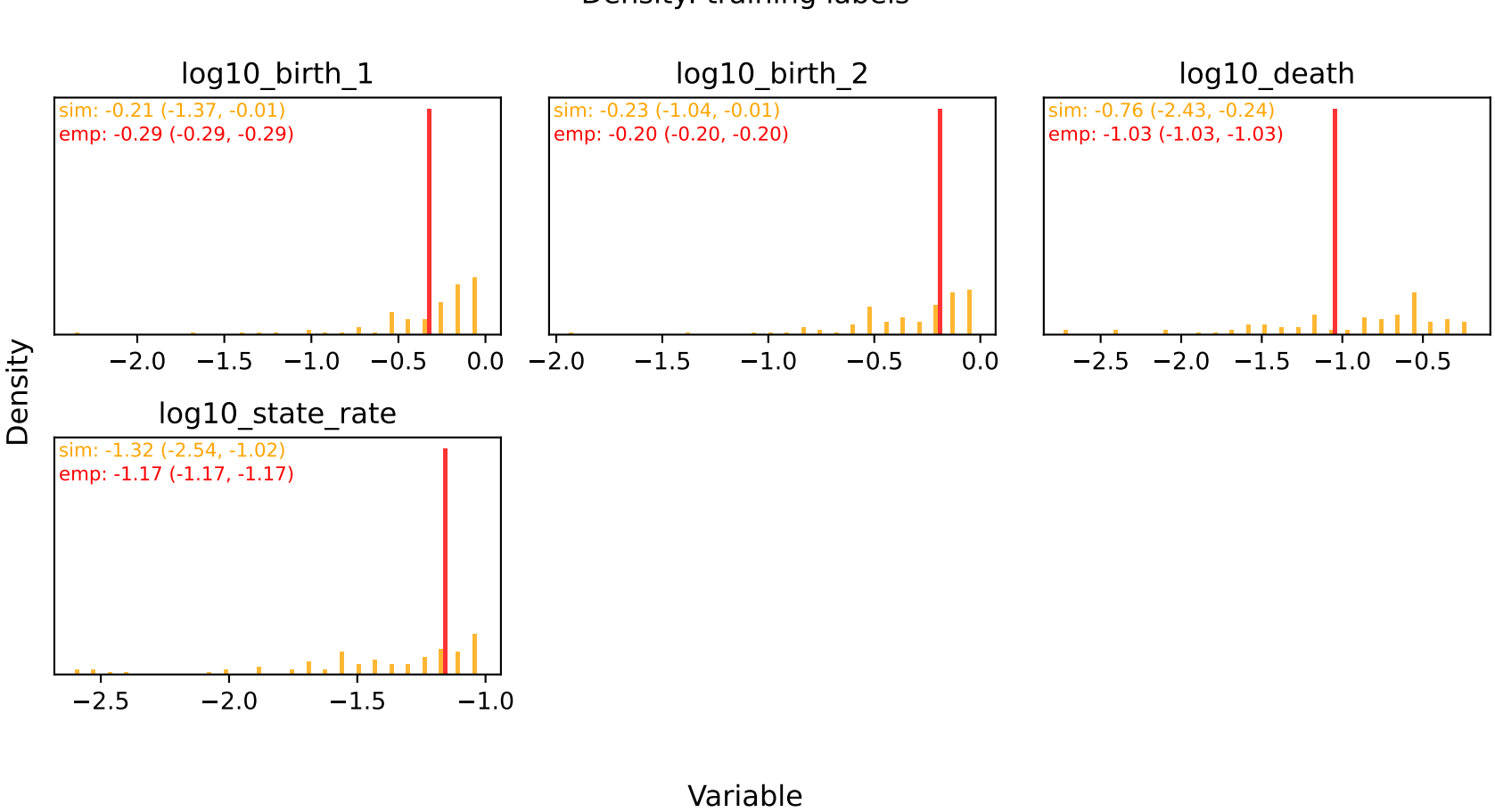

Estimate: out.empirical.0
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
−1.4
-2.64E-01
-2.92E-01
-3.20E-01
-1.17E-02
-1.98E-01
-3.84E-01
-9.25E-01
-1.03E+00
-1.13E+00
-9.80E-01
-1.17E+00
-1.37E+00
log10_birth_1
log10_birth_2
log10_death
log10_state_rate

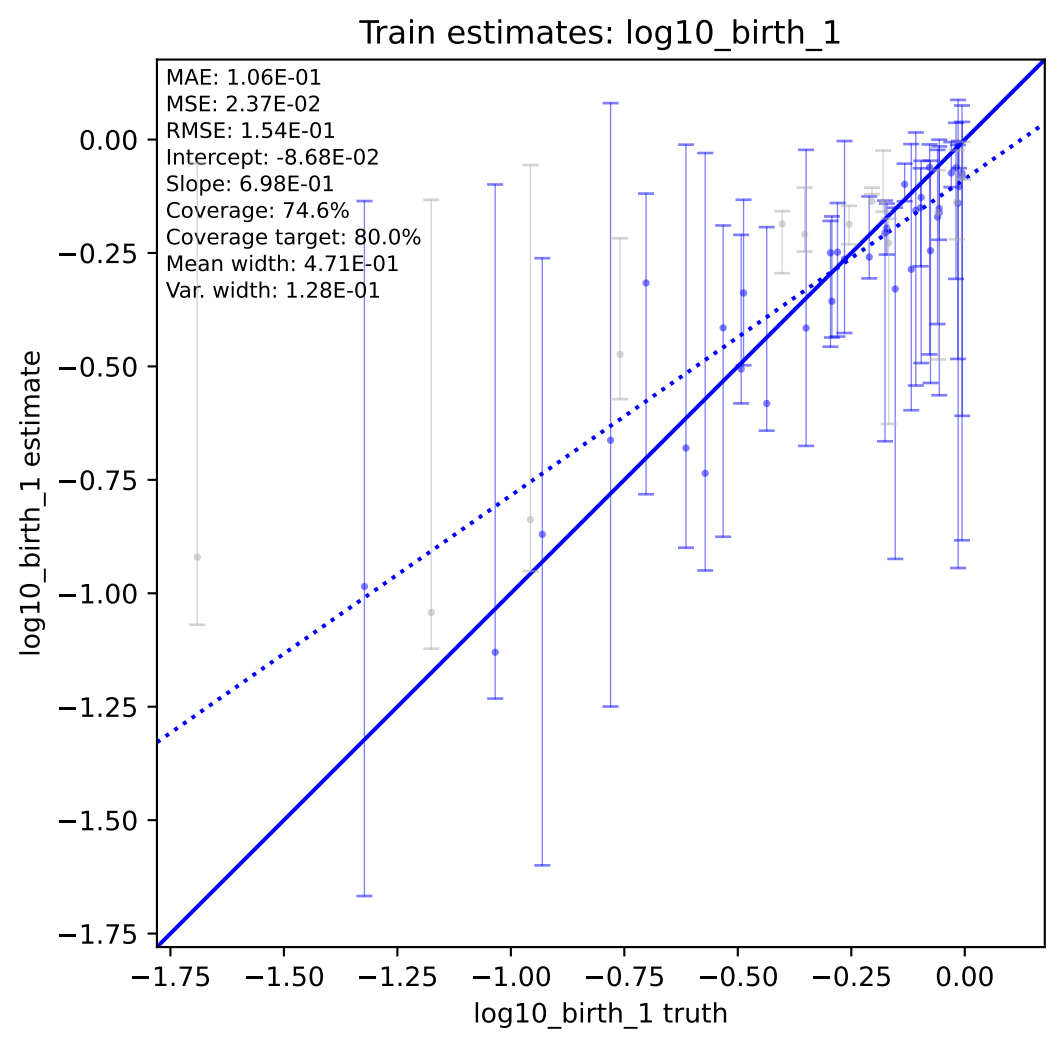

PCA: training aux. data
PC2 (31%)
% variance
PC
PC3 (15%)
PC4 (9%)
PC1 (38%)
PC2 (31%)
PC3 (15%)

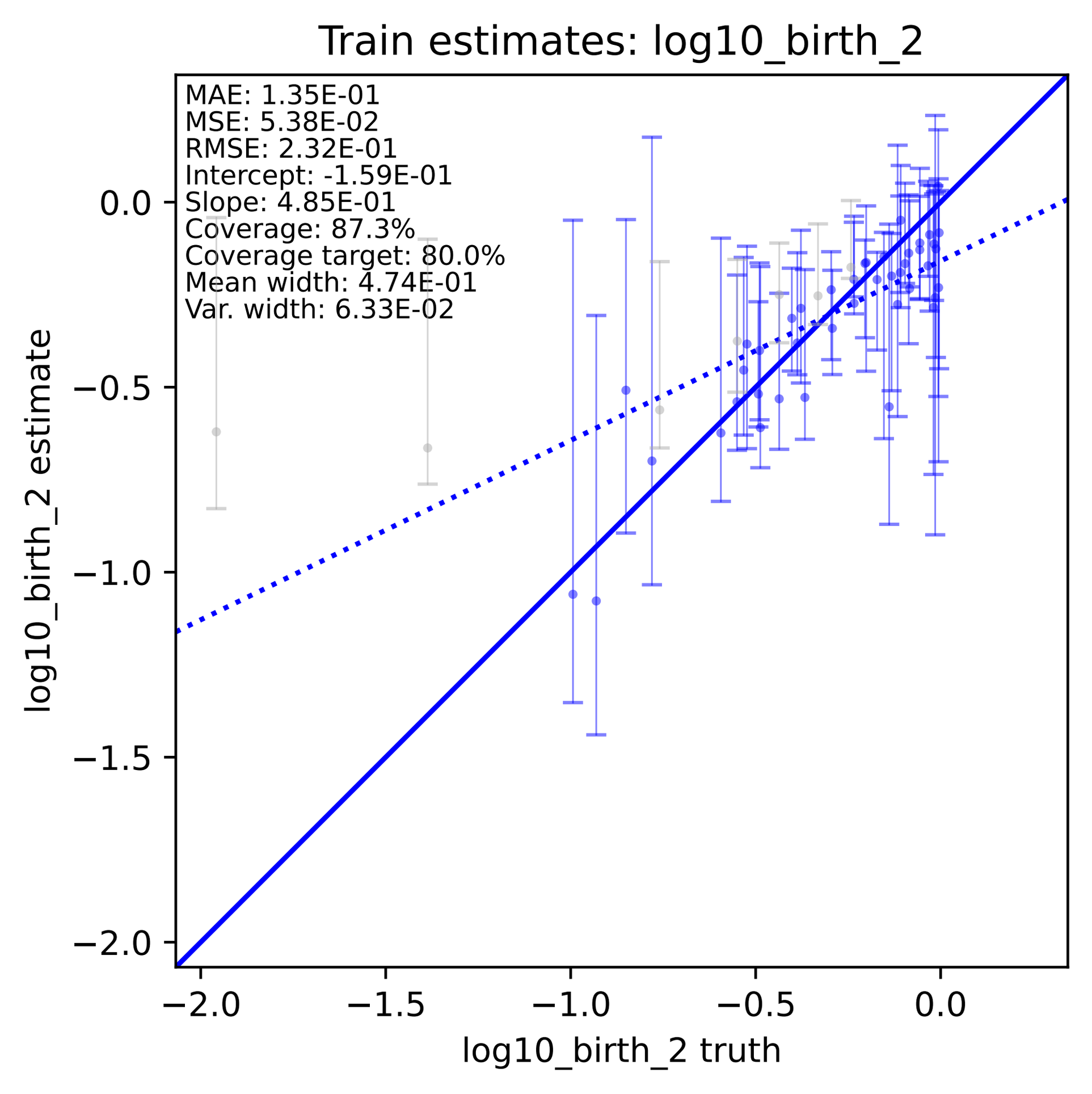

PCA: training labels
PC2 (25%)
% variance
PC
PC3 (18%)
PC4 (9%)
PC1 (46%)
PC2 (25%)
PC3 (18%)

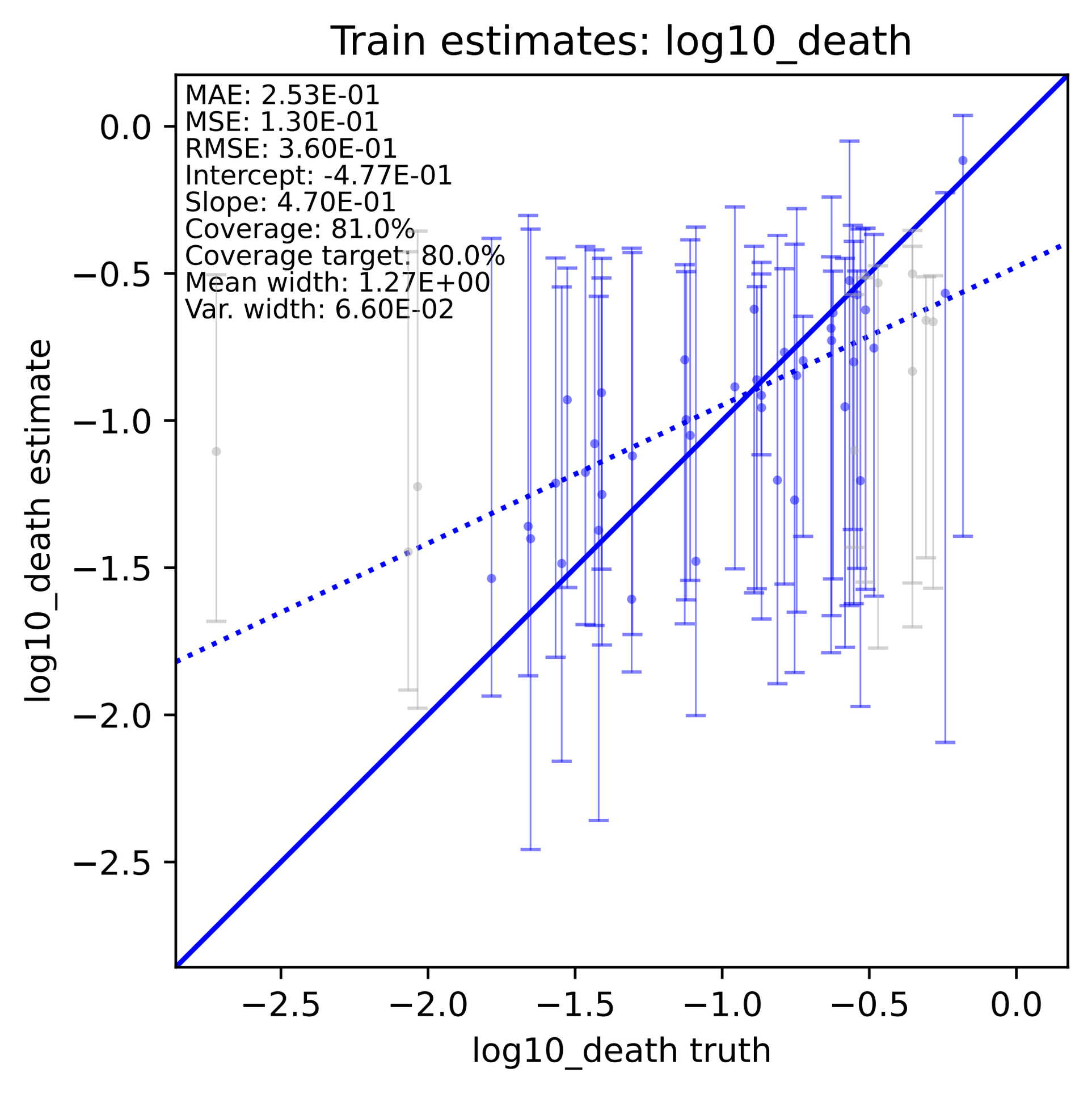

PCA: training phy. data
PC2 (5%)
% variance
PC
PC3 (2%)
PC4 (1%)
PC1 (33%)
PC2 (5%)
PC3 (2%)

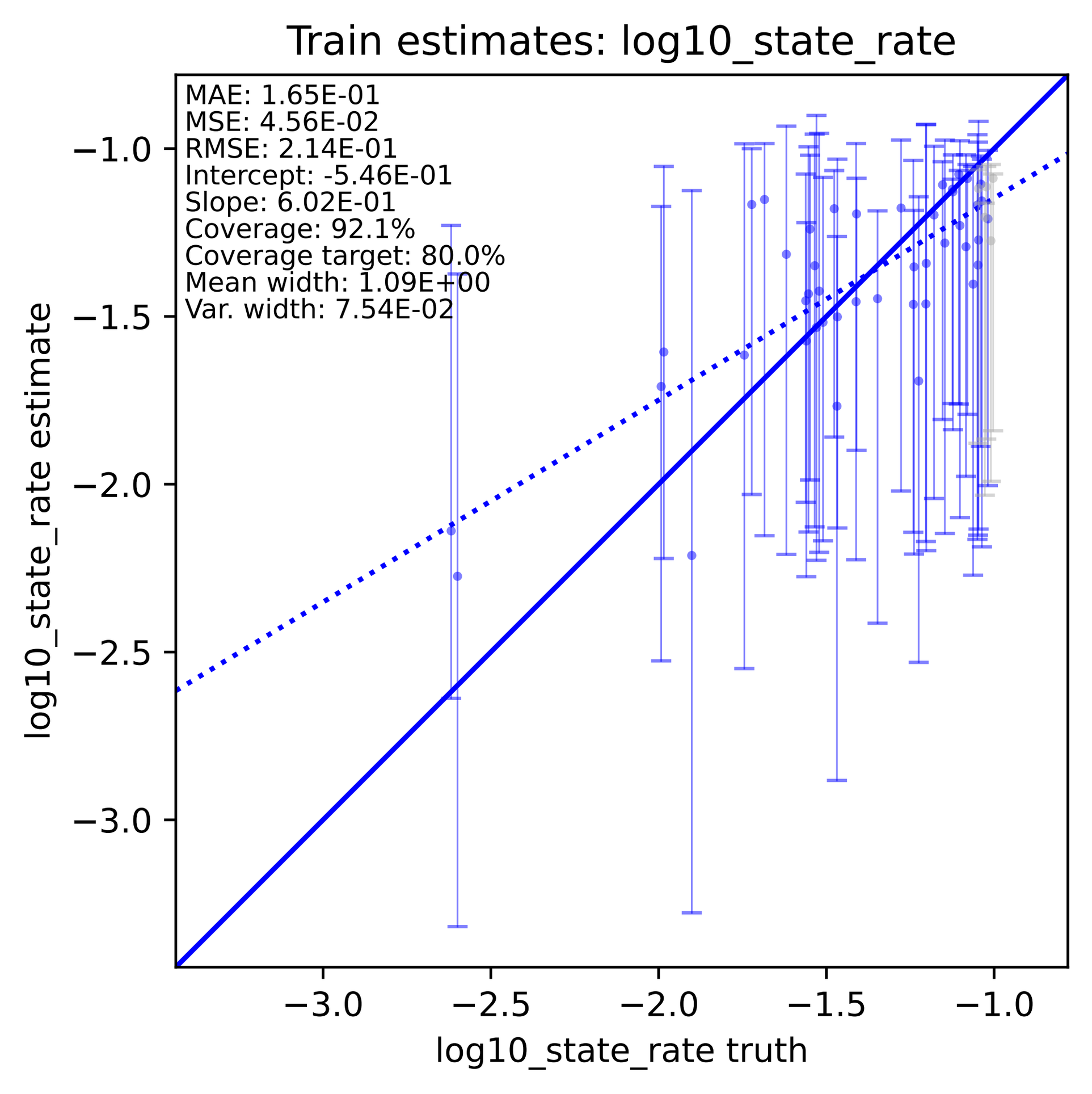

Density: training aux. data
N_bar
sim: 1.06 (0.68, 1.14)
emp: 1.06 (1.06, 1.06)
age_var
sim: 0.92 (0.16, 2.19)
emp: 0.70 (0.70, 0.70)
brlen_var
sim: 0.46 (-0.29, 1.68)
emp: 0.37 (0.37, 0.37)
colless
sim: -1.64 (-1.80, -0.47)
emp: -1.70 (-1.70, -1.70)
f_dat_0
sim: 0.52 (0.10, 1.00)
emp: 0.63 (0.63, 0.63)
f_dat_1
sim: 0.48 (0.03, 0.90)
emp: 0.37 (0.37, 0.37)
log10_B1
sim: 2.42 (0.82, 2.43)
emp: 2.42 (2.42, 2.42)
log10_age_mean
sim: 0.59 (0.08, 1.19)
emp: 0.59 (0.59, 0.59)
log10_brlen_mean
sim: 0.29 (-0.20, 0.91)
emp: 0.29 (0.29, 0.29)
log10_root_age
sim: 1.20 (0.74, 1.77)
emp: 1.04 (1.04, 1.04)
log10_sample_frac
sim: -0.70 (-0.99, 0.00)
emp: -0.95 (-0.95, -0.95)
log10_tree_length
sim: 3.20 (1.91, 3.71)
emp: 3.29 (3.29, 3.29)
n_dat_0
sim: 5.28 (2.08, 6.17)
emp: 5.76 (5.76, 5.76)
n_dat_1
sim: 5.24 (0.00, 6.11)
emp: 5.21 (5.21, 5.21)
num_taxa
sim: 500.00 (14.00, 500.00)
emp: 500.00 (500.00, 500.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.40 (-0.53, -0.24)
emp: -0.48 (-0.48, -0.48)
Density
Variable

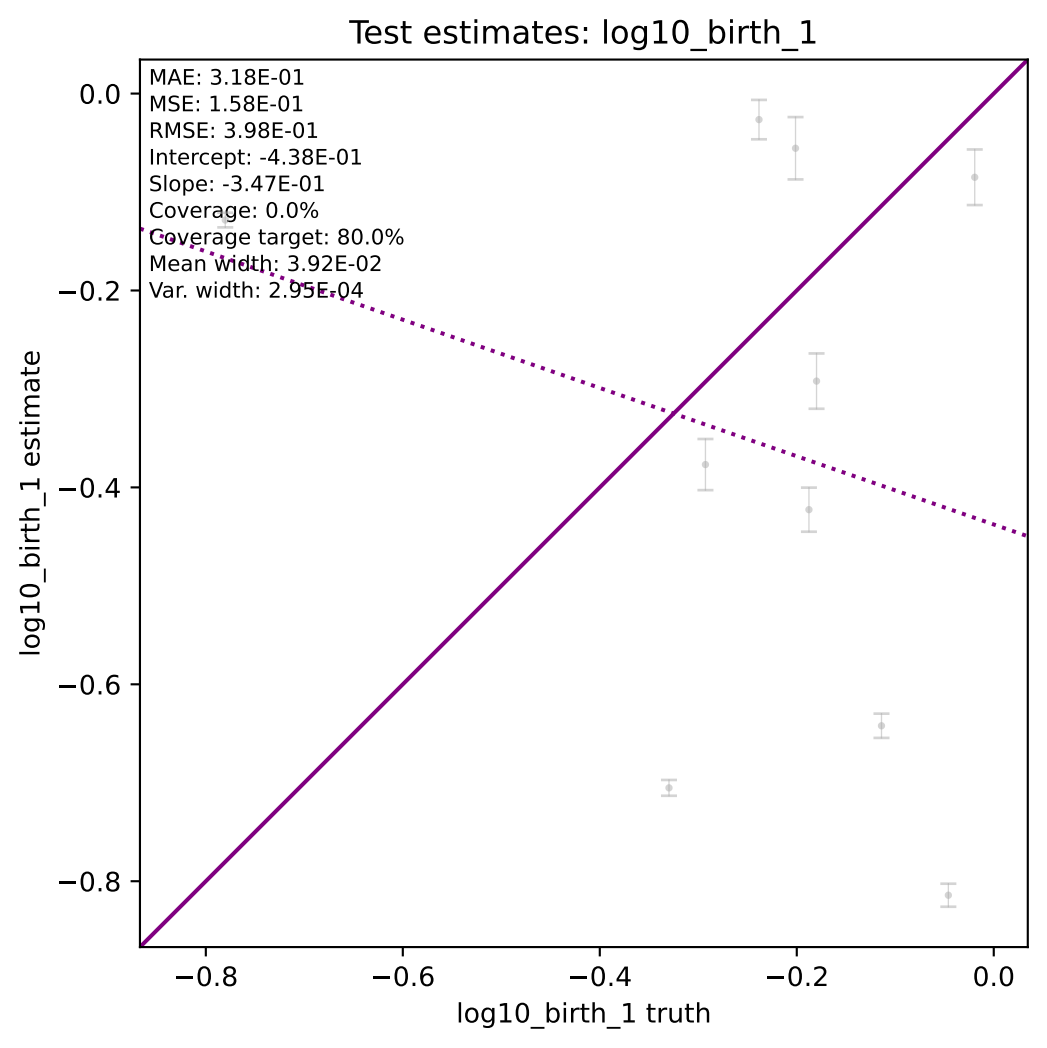

Density: training labels
log10_birth_1
sim: -0.21 (-1.37, -0.01)
emp: -0.29 (-0.29, -0.29)
log10_birth_2
sim: -0.23 (-1.04, -0.01)
emp: -0.20 (-0.20, -0.20)
log10_death
sim: -0.76 (-2.43, -0.24)
emp: -1.03 (-1.03, -1.03)
log10_state_rate
sim: -1.32 (-2.54, -1.02)
emp: -1.17 (-1.17, -1.17)
Density
Variable

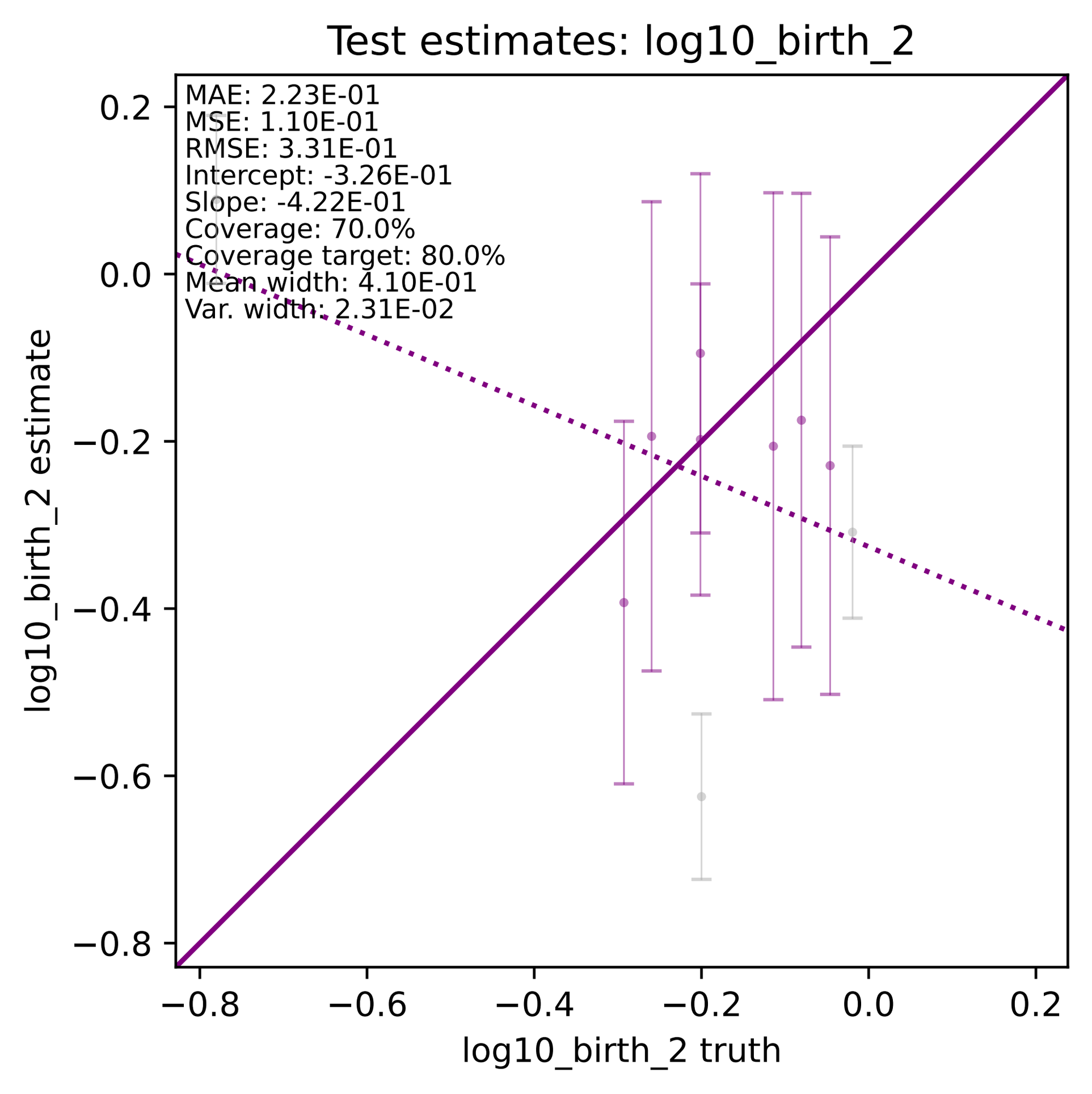

Train estimates: log10_birth_1
MAE: 1.06E-01
MSE: 2.37E-02
RMSE: 1.54E-01
Intercept: -8.68E-02
Slope: 6.98E-01
Coverage: 74.6%
Coverage target: 80.0%
Mean width: 4.71E-01
Var. width: 1.28E-01
log10_birth_1 estimate
log10_birth_1 truth
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75

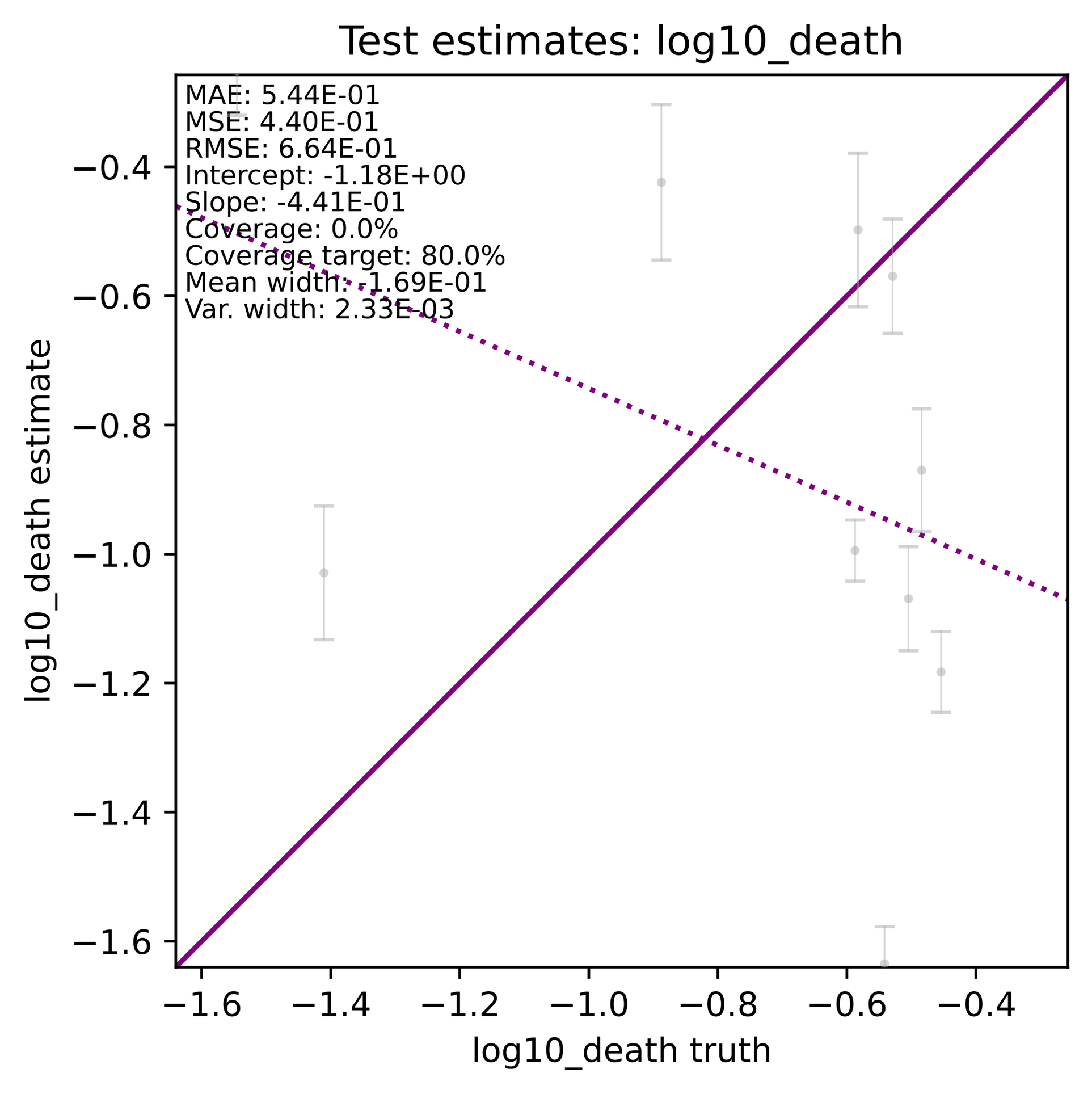

Train estimates: log10_birth_2
MAE: 1.35E-01
MSE: 5.38E-02
RMSE: 2.32E-01
Intercept: -1.59E-01
Slope: 4.85E-01
Coverage: 87.3%
Coverage target: 80.0%
Mean width: 4.74E-01
Var. width: 6.33E-02
log10_birth_2 estimate
log10_birth_2 truth
0.0
−0.5
−1.0
−1.5
−2.0
−2.0
−1.5
−1.0
−0.5
0.0

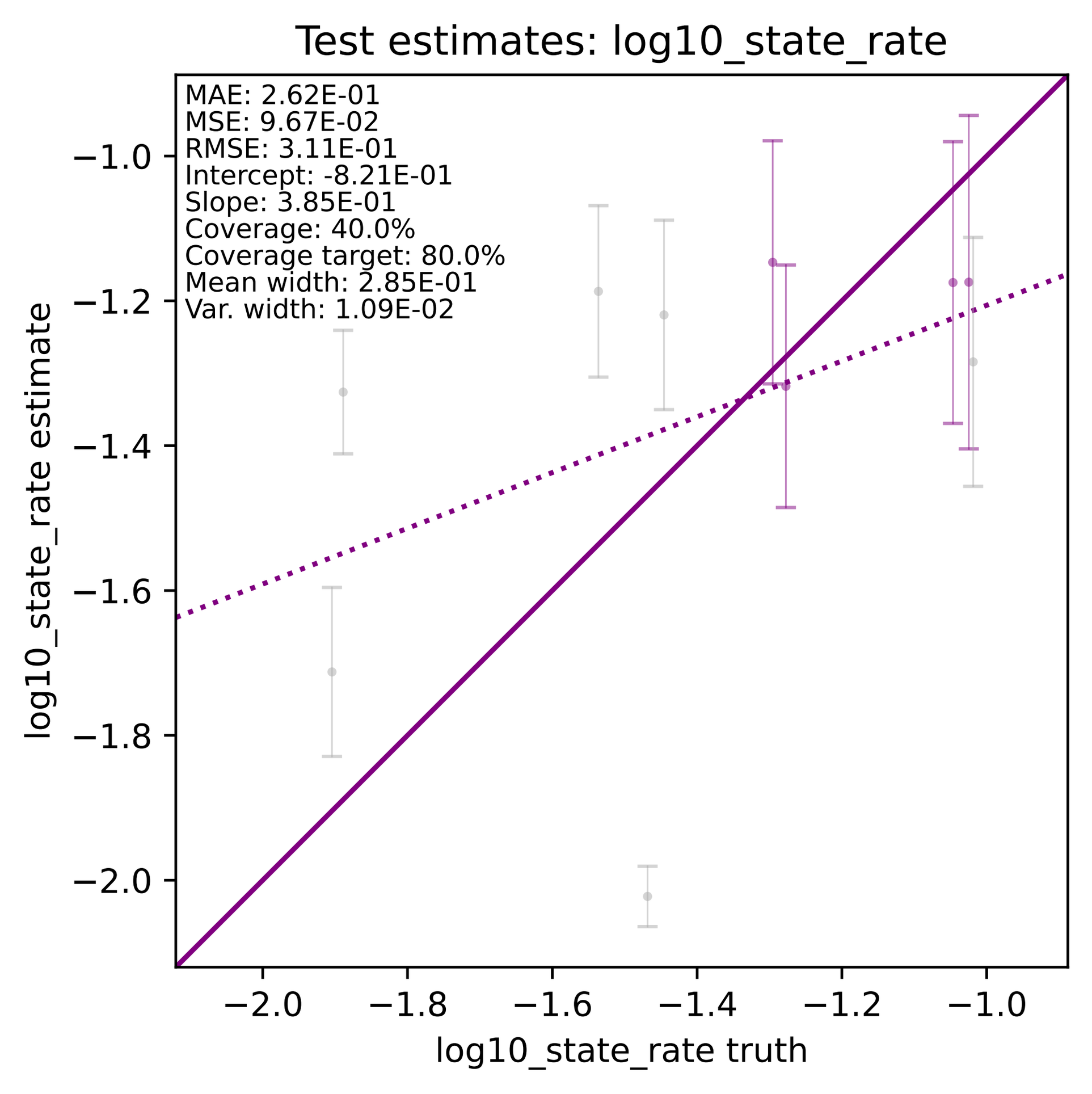

Train estimates: log10_death
MAE: 2.53E-01
MSE: 1.30E-01
RMSE: 3.60E-01
Intercept: -4.77E-01
Slope: 4.70E-01
Coverage: 81.0%
Coverage target: 80.0%
Mean width: 1.27E+00
Var. width: 6.60E-02
log10_death estimate
log10_death truth
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−2.5
−2.0
−1.5
−1.0
−0.5
0.0

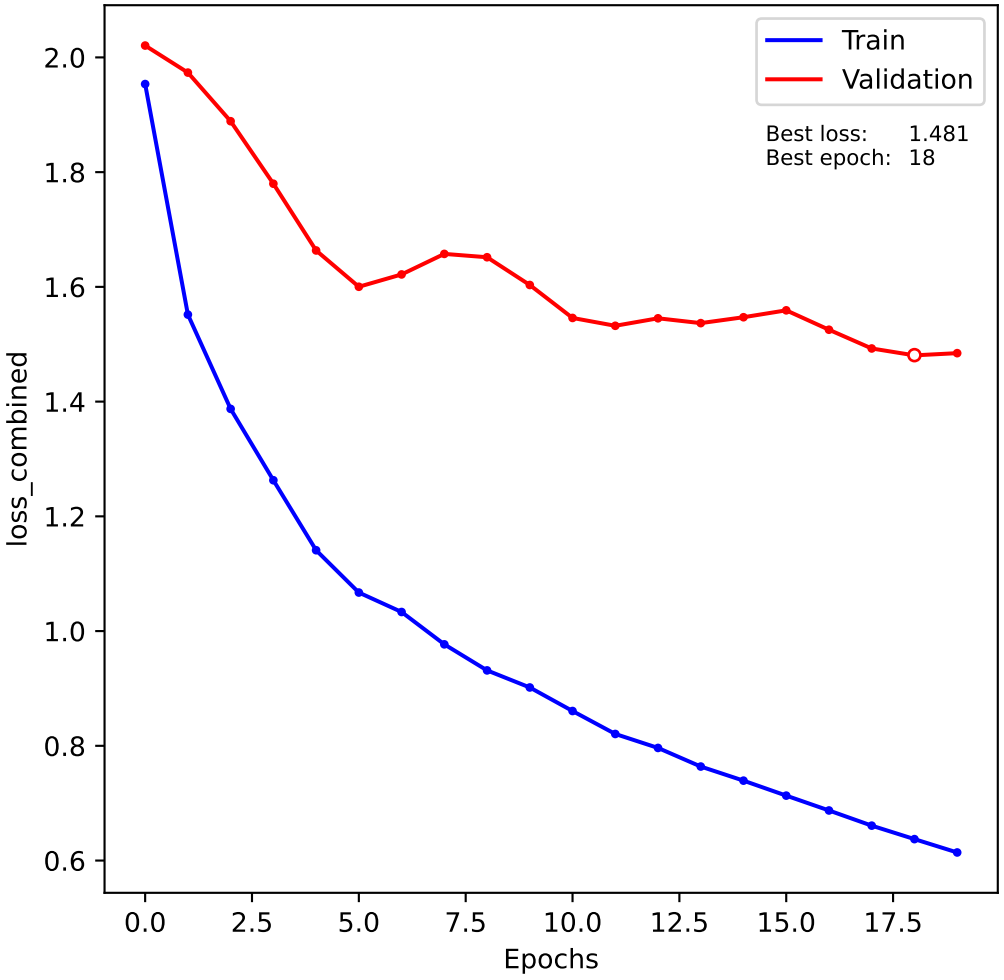

Train estimates: log10_state_rate
MAE: 1.65E-01
MSE: 4.56E-02
RMSE: 2.14E-01
Intercept: -5.46E-01
Slope: 6.02E-01
Coverage: 92.1%
Coverage target: 80.0%
Mean width: 1.09E+00
Var. width: 7.54E-02
log10_state_rate estimate
log10_state_rate truth
−1.0
−1.5
−2.0
−2.5
−3.0
−3.0
−2.5
−2.0
−1.5
−1.0

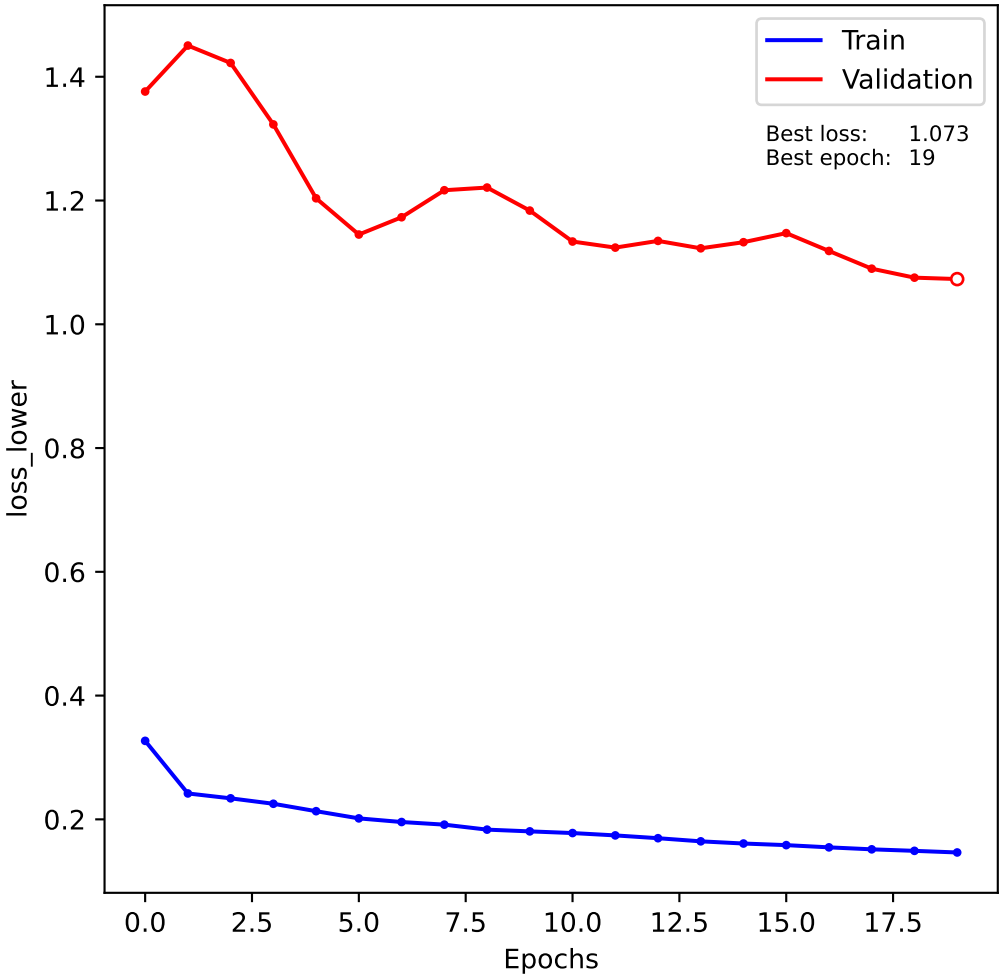

Test estimates: log10_birth_1
MAE: 3.18E-01
MSE: 1.58E-01
RMSE: 3.98E-01
Intercept: -4.38E-01
Slope: -3.47E-01
Coverage: 0.0%
Coverage target: 80.0%
Mean width: 3.92E-02
Var. width: 2.95E-04
log10_birth_1 estimate
log10_birth_1 truth
0.0
−0.2
−0.4
−0.6
−0.8
−0.8
−0.6
−0.4
−0.2
0.0

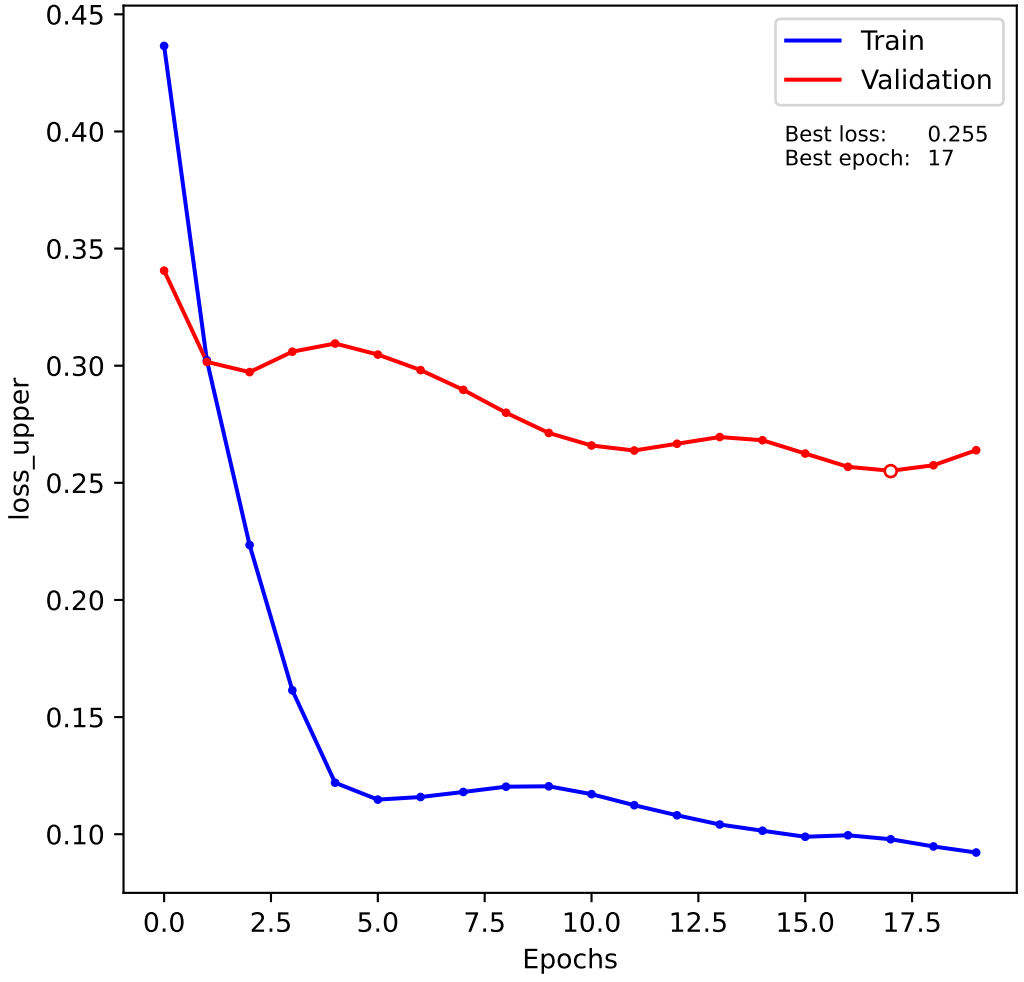

Test estimates: log10_birth_2
MAE: 2.23E-01
MSE: 1.10E-01
RMSE: 3.31E-01
Intercept: -3.26E-01
Slope: -4.22E-01
Coverage: 70.0%
Coverage target: 80.0%
Mean width: 4.10E-01
Var. width: 2.31E-02
log10_birth_2 estimate
log10_birth_2 truth
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−0.8
−0.6
−0.4
−0.2
0.0
0.2

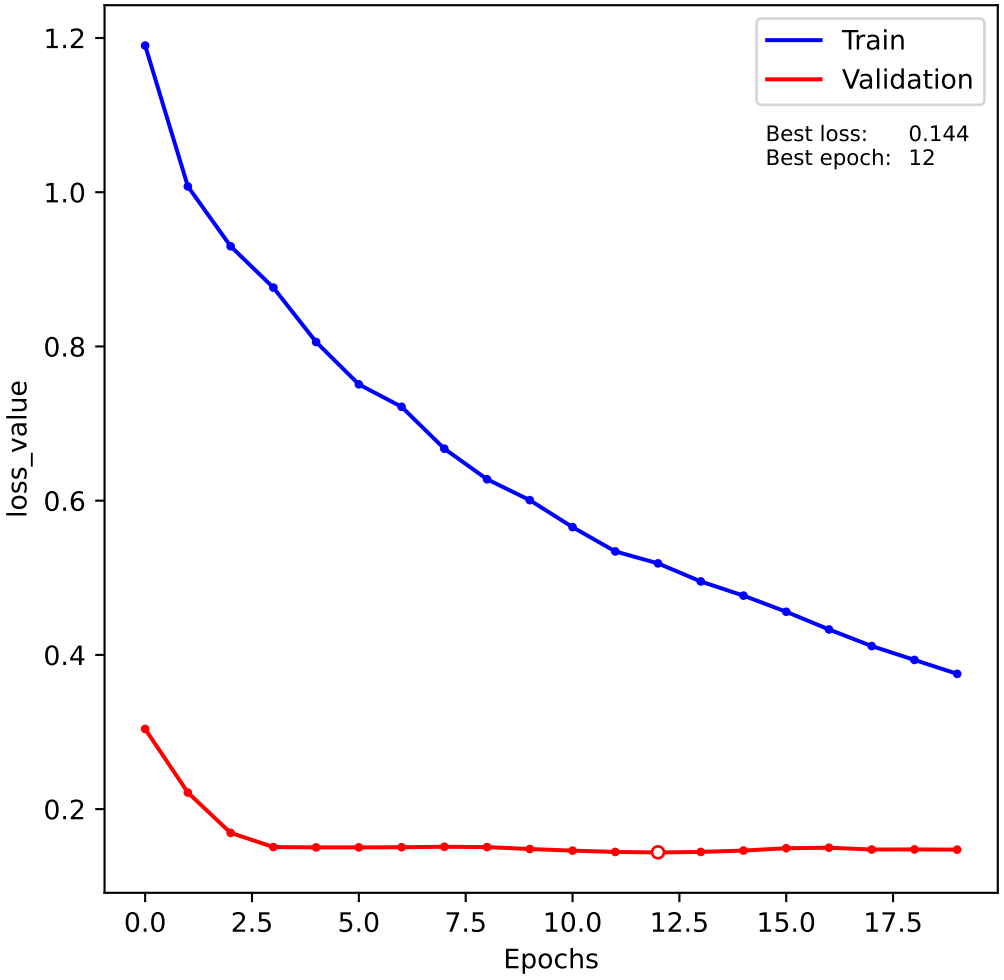

Test estimates: log10_death
MAE: 5.44E-01
MSE: 4.40E-01
RMSE: 6.64E-01
Intercept: -1.18E+00
Slope: -4.41E-01
Coverage: 0.0%
Coverage target: 80.0%
Mean width: -1.69E-01
Var. width: 2.33E-03
log10_death estimate
log10_death truth
−0.4
−0.6
−0.8
−1.0
−1.2
−1.4
−1.6

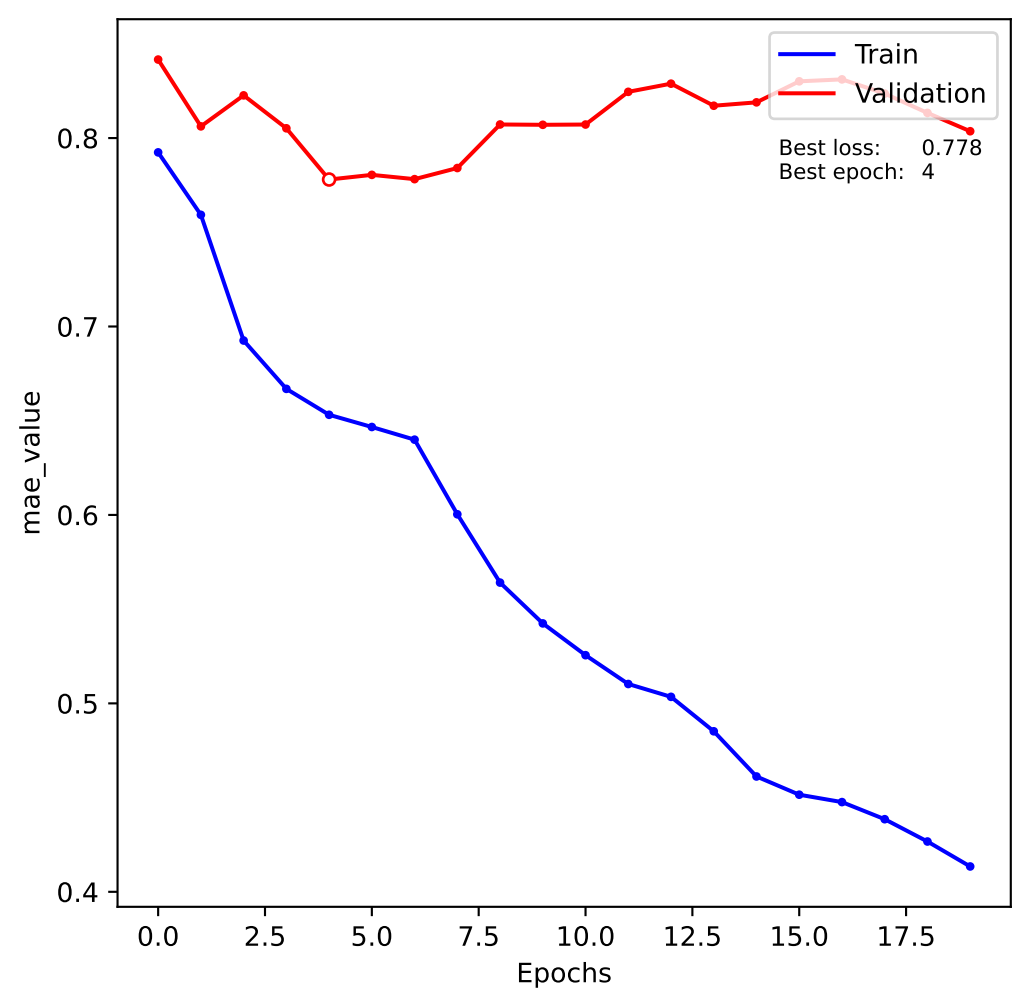

Test estimates: log10_state_rate
MAE: 2.62E-01
MSE: 9.67E-02
RMSE: 3.11E-01
Intercept: -8.21E-01
Slope: 3.85E-01
Coverage: 40.0%
Coverage target: 80.0%
Mean width: 2.85E-01
Var. width: 1.09E-02
log10_state_rate estimate
log10_state_rate truth
−1.0
−1.2
−1.4
−1.6
−1.8
−2.0
−2.0
−1.8
−1.6
−1.4
−1.2
−1.0

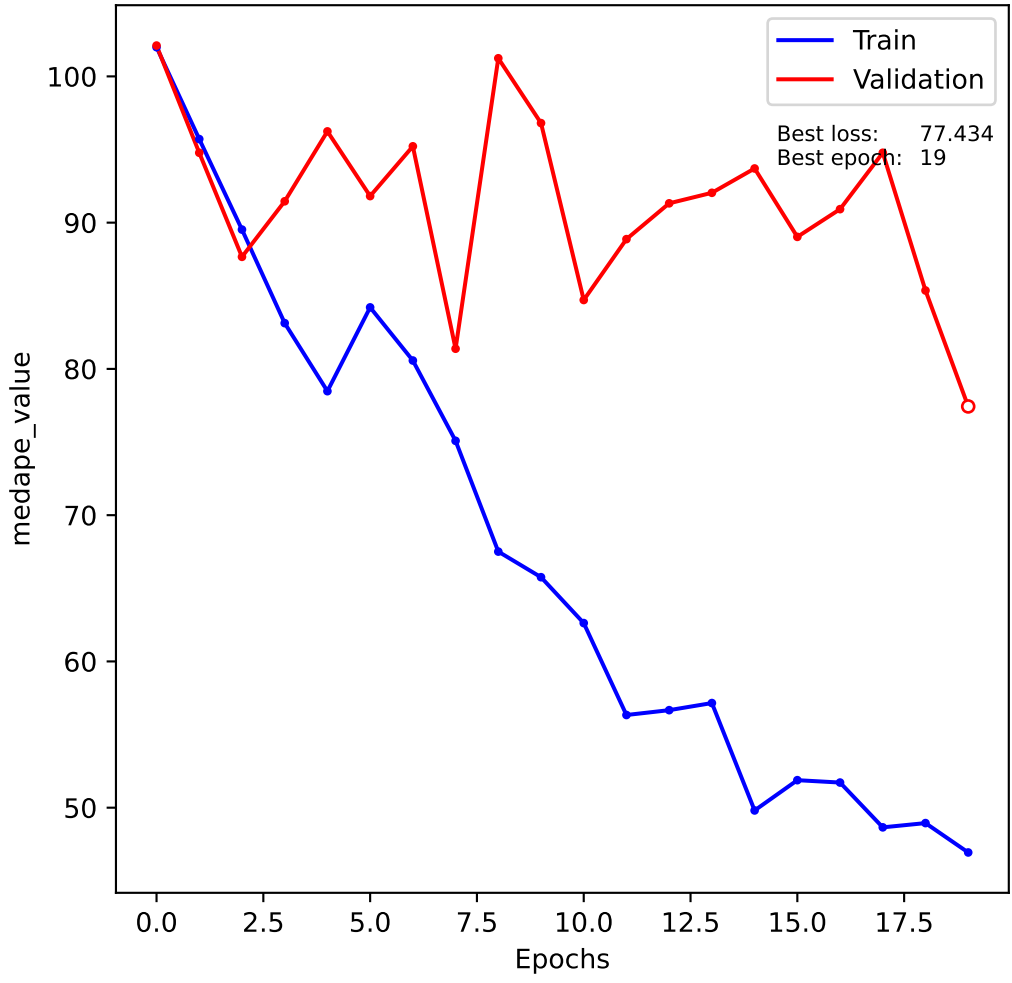

Train
Validation
Best loss: 1.481
Best epoch: 18
loss_combined
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Epochs

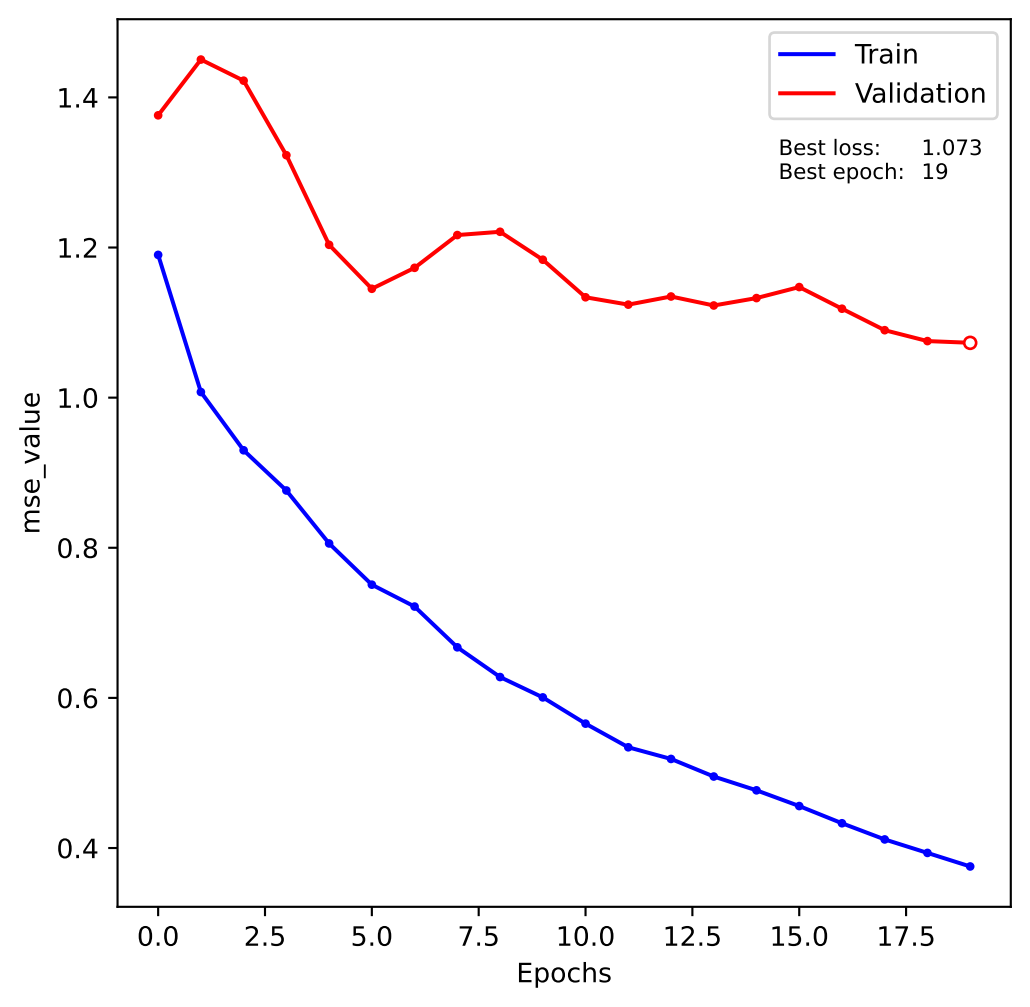

Train
Validation
Best loss: 1.073
Best epoch: 19
loss_lower
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Epochs

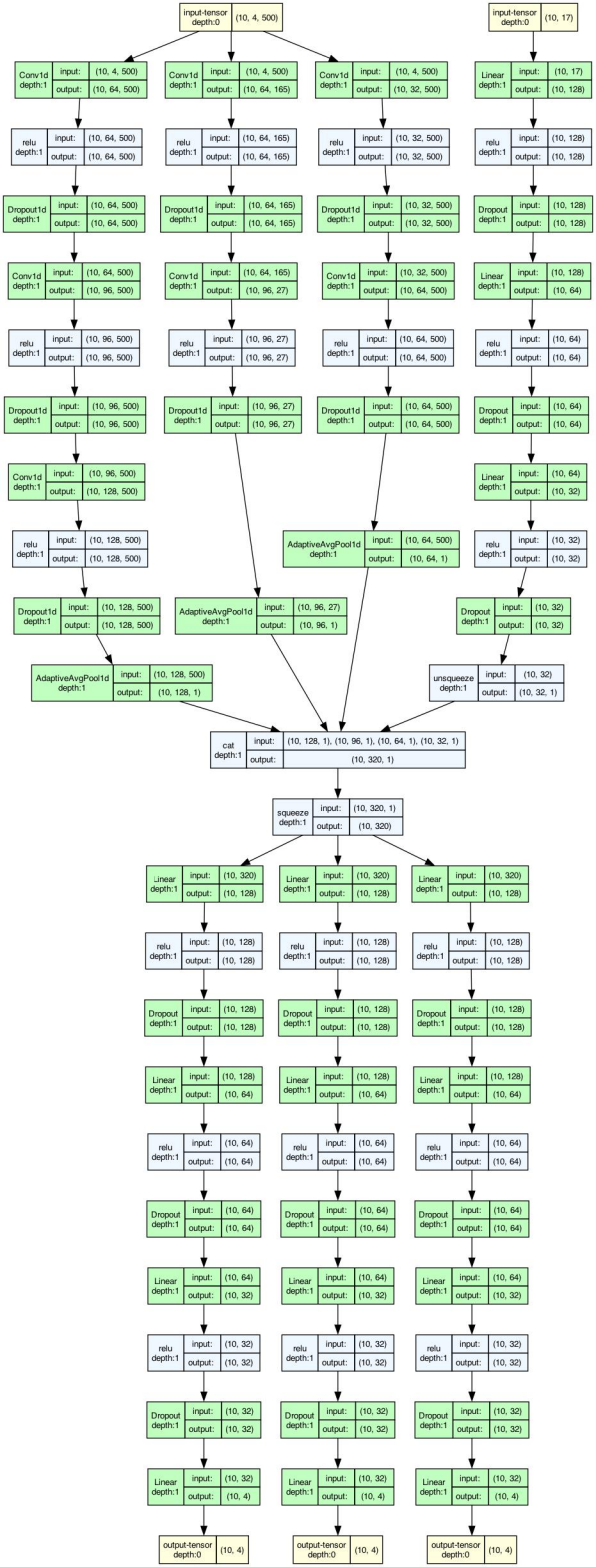

Train
Validation
Best loss: 0.255
Best epoch: 17
loss_upper
Epochs
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Train
Validation
Best loss: 0.144
Best epoch: 12
loss_value
Epochs
1.2
1.0
0.8
0.6
0.4
0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Train
Validation
Best loss: 0.778
Best epoch: 4
mae_value
0.8
0.7
0.6
0.5
0.4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Epochs


Train
Validation
Best loss: 77.434
Best epoch: 19
medape_value
Epochs
100
90
80
70
60
50
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Train
Validation
Best loss: 1.073
Best epoch: 19
mse_value
Epochs
1.4
1.2
1.0
0.8
0.6
0.4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


input-tensor depth:0 (10, 4, 500)
input-tensor depth:0 (10, 17)
Conv1d depth:1 input: (10, 4, 500) output: (10, 64, 500)
Conv1d depth:1 input: (10, 4, 500) output: (10, 64, 165)
Conv1d depth:1 input: (10, 4, 500) output: (10, 32, 500)
Linear depth:1 input: (10, 17) output: (10, 128)
relu depth:1 input: (10, 64, 500) output: (10, 64, 500)
relu depth:1 input: (10, 64, 165) output: (10, 64, 165)
relu depth:1 input: (10, 32, 500) output: (10, 32, 500)
relu depth:1 input: (10, 128) output: (10, 128)
Dropout1d depth:1 input: (10, 64, 500) output: (10, 64, 500)
Dropout1d depth:1 input: (10, 64, 165) output: (10, 64, 165)
Dropout1d depth:1 input: (10, 32, 500) output: (10, 32, 500)
Dropout depth:1 input: (10, 128) output: (10, 128)
Conv1d depth:1 input: (10, 64, 500) output: (10, 96, 500)
Conv1d depth:1 input: (10, 64, 165) output: (10, 96, 27)
Conv1d depth:1 input: (10, 32, 500) output: (10, 64, 500)
Linear depth:1 input: (10, 128) output: (10, 64)
relu depth:1 input: (10, 96, 500) output: (10, 96, 500)
relu depth:1 input: (10, 96, 27) output: (10, 96, 27)
relu depth:1 input: (10, 64, 500) output: (10, 64, 500)
relu depth:1 input: (10, 64) output: (10, 64)
Dropout1d depth:1 input: (10, 96, 500) output: (10, 96, 500)
Dropout1d depth:1 input: (10, 96, 27) output: (10, 96, 27)
Dropout1d depth:1 input: (10, 64, 500) output: (10, 64, 500)
Dropout depth:1 input: (10, 64) output: (10, 64)
Conv1d depth:1 input: (10, 96, 500) output: (10, 128, 500)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 128, 500) output: (10, 128, 500)
AdaptiveAvgPool1d depth:1 input: (10, 64, 500) output: (10, 64, 1)
relu depth:1 input: (10, 32) output: (10, 32)
Dropout1d depth:1 input: (10, 128, 500) output: (10, 128, 500)
AdaptiveAvgPool1d depth:1 input: (10, 96, 27) output: (10, 96, 1)
Dropout depth:1 input: (10, 32) output: (10, 32)
AdaptiveAvgPool1d depth:1 input: (10, 128, 500) output: (10, 128, 1)
unsqueeze depth:1 input: (10, 32) output: (10, 32, 1)
cat depth:1 input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1) output: (10, 320, 1)
squeeze depth:1 input: (10, 320, 1) output: (10, 320)
Linear depth:1 input: (10, 320) output: (10, 128)
Linear depth:1 input: (10, 320) output: (10, 128)
Linear depth:1 input: (10, 320) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
Dropout depth:1 input: (10, 128) output: (10, 128)
Dropout depth:1 input: (10, 128) output: (10, 128)
Dropout depth:1 input: (10, 128) output: (10, 128)
Linear depth:1 input: (10, 128) output: (10, 64)
Linear depth:1 input: (10, 128) output: (10, 64)
Linear depth:1 input: (10, 128) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
Dropout depth:1 input: (10, 64) output: (10, 64)
Dropout depth:1 input: (10, 64) output: (10, 64)
Dropout depth:1 input: (10, 64) output: (10, 64)
Linear depth:1 input: (10, 64) output: (10, 32)
Linear depth:1 input: (10, 64) output: (10, 32)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
Dropout depth:1 input: (10, 32) output: (10, 32)
Dropout depth:1 input: (10, 32) output: (10, 32)
Dropout depth:1 input: (10, 32) output: (10, 32)
Linear depth:1 input: (10, 32) output: (10, 4)
Linear depth:1 input: (10, 32) output: (10, 4)
Linear depth:1 input: (10, 32) output: (10, 4)
output-tensor depth:0 (10, 4)
output-tensor depth:0 (10, 4)
output-tensor depth:0 (10, 4)